

Data Flow Diagrams and User Stories

Automated Network Request Management

Context-Level Data Flow

Employee → Network Access Request Catalog Item → ServiceNow Flow Designer → Network Request Table → Approval & Assignment → Notification → Employee

Level-1 Data Flow

Step	Input	Process	Output
1. Request Submission	Employee enters request details	Employee submits the Network Access Request catalog item	Submitted request
2. Validation	Submitted request	Validate mandatory fields and catalog variables	Valid request / Validation message
3. Record Creation	Valid request	Flow Designer creates a Network Request record in the custom table	Network Request Record
4. Approval	Network Request Record	Route request to Manager or Network Administrator for approval	Approved / Rejected Request
5. Task Assignment	Approved request	Automatically assign request to the appropriate IT Support Group	Assigned Network Task
6. Notification	Assignment and request status	Send automated notifications to requester and support team	Email / System Notification
7. Monitoring	Request records	Generate reports and dashboards to monitor request progress	Reports and Dashboards

User Stories

ID	As a...	I want to...	So that...
US-01	Employee	submit a network access request through the Service Catalog	I can request network services easily.
US-02	Employee	select the required request type, device type, and access level	my request contains accurate information.

US-03	Employee	provide a business justification and requested user details	the approver understands why access is required.
US-04	Employee	receive confirmation after submitting my request	I know my request has been successfully created.
US-05	Employee	track the status of my network request	I know whether my request is pending, approved, or completed.
US-06	Network Administrator	review and approve network requests	only authorized requests are processed.
US-07	IT Support Engineer	receive assigned network requests automatically	I can complete requests without manual assignment.
US-08	Administrator	view reports and dashboards	I can monitor request volume, approvals, and performance.

Proposed Flow Diagram Description

Employee submits a **Network Access Request** through the **Service Catalog** → **Flow Designer** retrieves the submitted catalog variables → validates the request information → creates a **Network Request Record** in the custom table → routes the request for approval → automatically assigns the approved request to the appropriate IT support group → sends notification emails to the requester and support team → updates reports and dashboards for monitoring and tracking.

Exception Flows

- **Mandatory field missing:** Display a validation message and prevent request submission.
- **Invalid access level selected:** Reject the request and ask the user to provide valid information.
- **Flow execution failure:** Log the error and notify the administrator for investigation.
- **Approval rejected:** Update the request status to **Rejected** and notify the requester.
- **Assignment group unavailable:** Place the request in an exception queue for manual assignment.
- **Notification failure:** Keep the request active and retry notification or allow manual follow-up.
- **Record creation failure:** Do not create duplicate records; log the failure and notify the administrator.